

DM I

Block 'Clustering', Unit 'Cluster Evaluation', Replacement for slide 56/71

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Internal indices: Silhouette coefficient

(Rousseeuw, 1987)²

For a clustering ζ over a set D , a distance function $d()$ ¹ and a cluster $X \in \zeta$:

Silhouette of a data point $x \in D$

Rousseeuw (1987)

- ▶ $a(x) = \frac{\sum_{u \in X \setminus \{x\}} d(x, u)}{|X| - 1}$ – average distance of x to the other members of 'its' cluster
- ▶ $bb(x, Y) = \frac{\sum_{u \in Y} d(x, u)}{|Y|}$ – average distance of x to the members of cluster $Y \in \zeta \setminus X$,
and then $b(x) = \min_{Y \in \zeta \setminus \{X\}} bb(x, Y)$ ^a
- ▶ $silhouette(x) = \begin{cases} 1 - \frac{a(x)}{b(x)} & \text{if } a(x) < b(x) \\ \frac{b(x)}{a(x)} - 1 & \text{if } a(x) \geq b(x) \end{cases}$

^a $Y_0 = \arg \min_{Y \in \zeta \setminus \{X\}} bb(x, Y)$ is the cluster to which we would have placed x as second-best choice, i.e. if X were not there. This intuition comes from Rousseeuw (1987), page 56.

Aggregations

[Ch 7, page 582]

$$cluSilh(X) = \frac{\sum_{x \in X} silhouette(x)}{|X|}$$

$$Silh(\zeta) = \frac{\sum_{X \in \zeta} cluSilh(X)}{|\zeta|}$$

¹Normalization has already been done, hence $d(x, u) \in [0, 1]$ for each $x, u \in D$.

²Rousseeuw, P. J. (1987) 'Silhouettes: a graphical aid to the interpretation and validation of cluster analysis' Journal of Computational and Applied Mathematics, 20, 53-65. DOI: /10.1016/0377-0427(87)90125-7

Questions on the Silhouette coefficient

- ① In the denominator of $bb()$ we see $|Y|$ while in the denominator of $a()$ we see $|X|$. Is this a typo? Justify your answer.
- ② Can we skip the function $bb()$ by defining $b(x) = \min_{Y \in \zeta \setminus \{X\}} \frac{\sum_{u \in Y} d(x, u)}{|Y|}$? Justify your answer.
- ③ We define a function $f(x, Z) = \frac{\sum_{u \in Z} d(x, u)}{|Z|}$. Can we then say that:
 - ▶ $a(x) = f(x, X)$ where X is the cluster to which x belongs.
 - ▶ $bb(x, Y) = f(x, Y)$ where Y is a cluster that does not contain x .Justify your answers. If you answered No to one of these questions, how can you modify the equation, so that it fits?
- ④ Rousseeuw (1987) wrote: 'this is like the second-best choice for object i : if it could not be accommodated into cluster A , which cluster B would be the closest competitor?'³ However, this looks a bit unfair given the definition of $a(x)$, because of the denominator $|X| - 1$. Should we use $|X|$ instead in $a(x)$?

³In the notation here, x stands for i , A for X and B for $Y_0 = \arg \min_{Y \in \zeta \setminus \{X\}} bb(x, Y)$.